

Civics Group	Index Number	Name (use BLOCK LETTERS)

H2



**ST. ANDREW'S JUNIOR COLLEGE
2013 JC2 Preliminary Examinations**

H2 BIOLOGY

9648/1

Paper 1: Multiple Choice

Friday

20 September 2013

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **40** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice answer sheet.

INFORMATION TO CANDIDATES

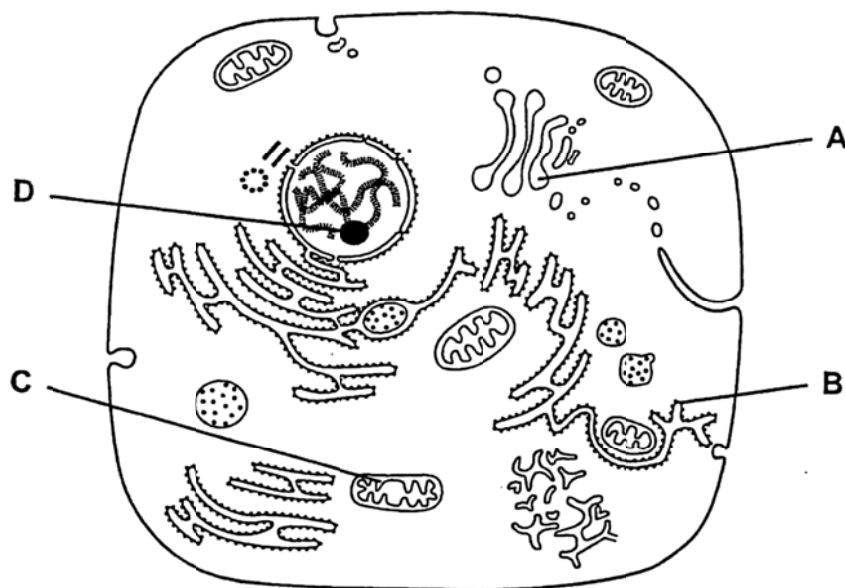
Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit the multiple choice answer sheet only.

This document consists of **15** printed pages.

[Turn over

- 1 The nuclei of both plant and animal cells contain one or more dense bodies known as nucleoli. Which one of the following correctly describes the function of nucleoli?
- A The formation of new DNA molecules.
 - B The organization of the spindle during nuclear division.
 - C Synthesis of chromatin fibres.
 - D The formation of ribosomes.
- 2 What could be a possible explanation for the ability of lysosomes to withstand self-digestion?
- A lysosomes contain inactive hydrolytic enzymes
 - B lysosome membrane has numerous modified proteins with carbohydrate side-chains
 - C lysosomes do not contain lipases which are the enzymes capable of digesting the lipid membrane
 - D hydrolases in the lysosomes are inhibited by the acidic internal environment of the lysosomes
- 3 The diagram below shows the ultrastructure of a eukaryotic cell. Which organelle does not contain nucleic acid?



- 4 Some organisms change the composition of their membranes to maintain membrane fluidity when temperature changes.

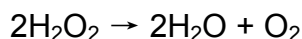
Which changes occur in the composition of the membranes following a change to low temperature?

- A Increase phospholipids, decrease cholesterol.
- B Increase saturated fatty acids, increase cholesterol.
- C Increase saturated fatty acids, decrease cholesterol.
- D Increase unsaturated fatty acids, increase cholesterol.

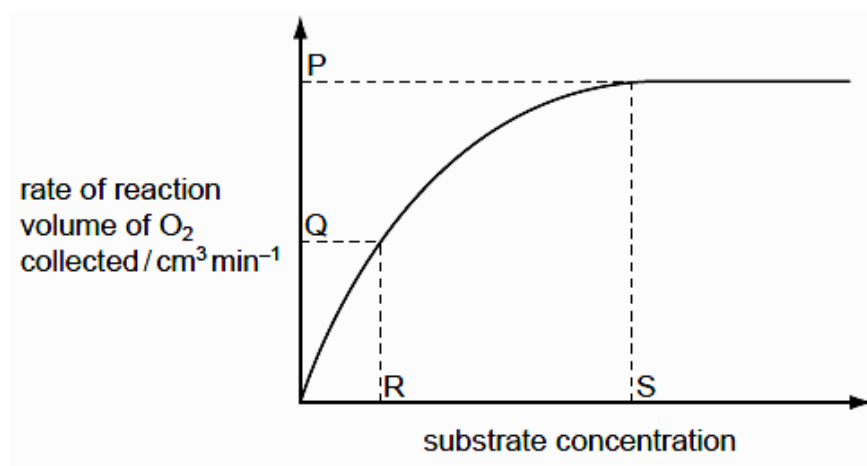
5 What does a haemoglobin molecule contain?

- A four iron (Fe^{2+}) ions attached to each haem group
- B four oxygen molecules attached to each haem group
- C four polypeptide chains each with one attached haem group
- D four polypeptide chains each with four attached haem groups

6 Liver tissue produces an enzyme called catalase which breaks down hydrogen peroxide into water and oxygen.



The rate of this reaction can be determined by measuring the volume of oxygen produced in a given length of time. Students added small cubes of fresh liver tissue to a range of hydrogen peroxide solutions and measured the volumes of oxygen produced. Their data were used to produce the graph showing how changing the concentration of hydrogen peroxide affected the rate of oxygen production.

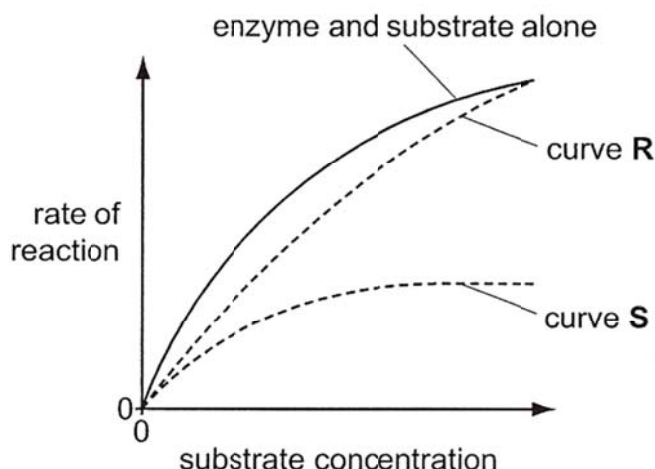


Which statements are correct?

- 1 At P, the rate of reaction is limited by the concentration of enzyme.
- 2 At Q, all of the enzyme active sites are occupied by substrate molecules.
- 3 At Q, the rate of reaction is limited by the concentration of the substrate.
- 4 At S, all of the enzyme active sites are occupied by substrate molecules.

- A 1 and 4
- B 2 and 3
- C 2 and 4
- D 1, 3 and 4

- 7 The graph shows the effect of two different compounds on the rate of reaction of the enzyme at different substrate concentrations.



Which statement is true?

- A Compound which results in curve R is a competitive inhibitor which also increases the K_m of the enzyme.
 - B Compound which results in curve R is a non-competitive inhibitor which did not affect the K_m of the enzyme.
 - C Compound which results in curve S is a competitive inhibitor which also decreases the K_m of the enzyme.
 - D Compound which results in curve S is a non-competitive inhibitor which increases the K_m of the enzyme.
- 8 A hypothetical diploid cell has four chromosomes and total amount of DNA in the cell is denoted by "X" before DNA replication.

Which of the following is correct?

	Cell Cycle Phase	Number of chromosomes per cell	Amount of DNA per cell
A	Interphase	8	2X
B	Telophase of mitosis	8	X
C	Metaphase of meiosis I	4	X
D	Anaphase of meiosis II	4	X

- 9 When identical twins marry identical twins, the children of both couple are genetically
- A identical unless crossing over takes place
 - B identical because of the effect of semi-conservative DNA replication prior to nuclear division
 - C different because non-disjunction of chromatids occurs
 - D different because of random segregation during parental meiosis

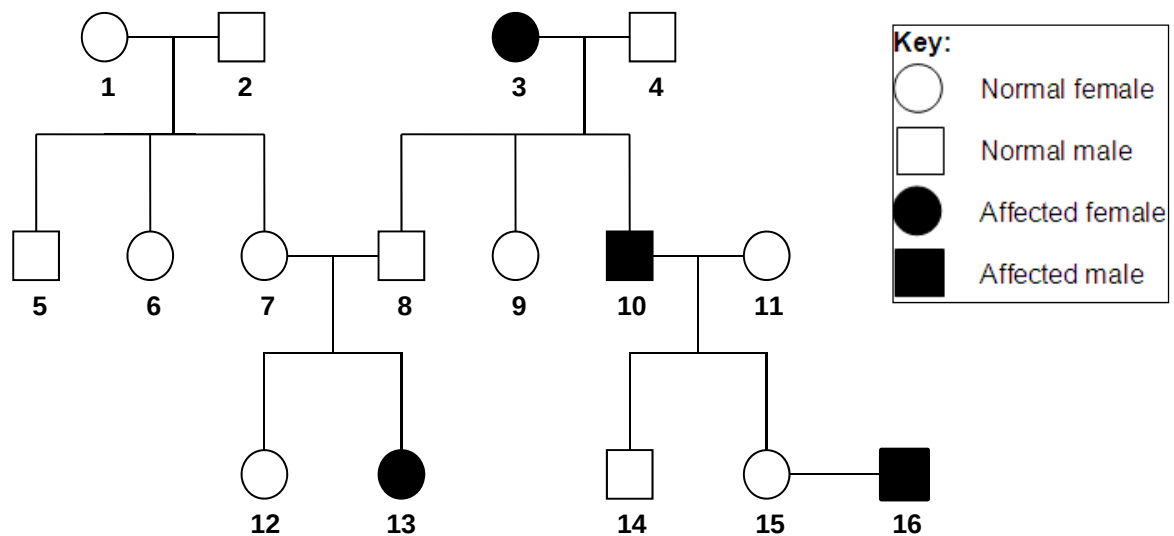
- 10** To function as the heritable genetic code, DNA molecules must have all of the following structural features except
- A** the ability to form complementary base-pairs with RNA molecules.
 - B** exist as a very stable, double helical form when being replicated
 - C** a sequence of nucleotides that can be decoded into a sequence of amino acids in a protein
 - D** ability to undergo semi-conservative replication
- 11** A particular eukaryotic protein is 300 amino acids long. Which of the following could be the minimum number of nucleotides in the DNA molecule containing the gene that codes for the amino acids in this protein?
- A** 100
 - B** 300
 - C** 900
 - D** 1800
- 12** Which process in bacteria always allows chromosomal and non-chromosomal DNA to be transferred?
- A** Binary fission.
 - B** Conjugation.
 - C** Transduction.
 - D** Transformation.
- 13** Which of the following statements about binary fission is not true?
- 1 The mode of replication is semi-conservative
 - 2 Chromosomes are separated to opposite ends of the cell.
 - 3 DNA replicates using the rolling circle mechanism.
 - 4 DNA replication at the *ori* is bi-directional.
- A** 3 only
 - B** 1 and 2
 - C** 2 and 4
 - D** 3 and 4
- 14** Which of the following statements concerning *lac* operon is true?
- 1 Transcription of *lac* operon takes place all the time.
 - 2 There is one single mRNA transcribed from the *lac* operon.
 - 3 There is one start and one stop codon in the mRNA of *lac* operon.
 - 4 The repressor molecule binds to the operator to turn off *lac* operon.
- A** 4 only
 - B** 1 and 3
 - C** 2 and 4
 - D** 2, 3 and 4

15 How do viruses cause diseases in animals?

- 1 They inhibit normal synthesis of host cell DNA, RNA, or protein.
- 2 They degrade the host cell's chromosomes.
- 3 They disrupt the oncogenes of the host cell causing uncontrolled cell division.
- 4 Their viral proteins and glycoproteins on the surface membrane of host cells cause them to be recognized and destroyed by the body's immune defences.

- A** 1 and 3
B 2 and 3
C 1, 2 and 4
D 1, 2, 3 and 4

16 The pedigree shows the inheritance of a condition known as galactosemia.



What is the probability that the first child of individuals 15 and 16 will be a normal boy?

- A** 0.125
B 0.25
C 0.50
D 0.75

- 17** Agouti mice have banded hairs, giving a grey colour. Black mice have unbanded hairs. White mice have no pigment. A cross between a homozygous black mouse and a white mouse produced offspring with agouti hair. Another cross between the black mouse and another white mouse produced some offspring with agouti hair and some with black hair.

What explains these observations on the phenotype of hair of mice?

- A** There is a single gene with two codominant alleles, black and white.
 - B** There is a single gene with three alleles in a dominance series, black → grey → white.
 - C** There are two epistatic genes, one controlling pigment production and one controlling banding.
 - D** There are two linked genes, one controlling pigment production and one controlling banding.
- 18** With reference to prokaryotic and eukaryotic genomes, which of the following statement is not true?
- A** They both have extrachromosomal DNA.
 - B** They both have non-coding regions.
 - C** They both have regulatory sequences.
 - D** They are both associated with histones.
- 19** Four different genes are regulated in different ways.

Gene 1 undergoes tissue-specific patterns of alternative splicing

Gene 2 is part of a group of structural genes controlled by the same regulatory sequences

Gene 3 is in some circumstances subject to methylation

Gene 4 codes for a repressor protein which acts at an operator site close by

Which role of the table correctly identifies which genes are prokaryotic and which are eukaryotic?

	prokaryotic	eukaryotic
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 3	1 and 4
D	2 and 4	1 and 3

- 20** Which of the following statements about the eukaryotic control elements are correct?
- 1 Attachment of the RNA polymerase II at the TATA box is achieved with the help of a series of specific transcription factors
 - 2 Upstream regulatory elements (UREs) increase the basal level of transcription
 - 3 A given gene may have multiple enhancers, each active at a different time or in a different cell type or location in the organism
 - 4 Repressors bind to silencer regions of DNA far upstream of promoters to repress transcription
- A** 1 and 4
B 2 and 3
C 2, 3 and 4
D 1, 2, 3 and 4

- 21** Cells taken from a human bone cancer multiplied readily in culture. Analysis showed that the cells were homozygous for the deletion of an allele on chromosome 13 coding for a protein, RB.

Addition of RB to these cells reduced their rate of division.

Which individual is **not** able to produce a tumour suppressor?

- A** Both chromosomes 13 carry alleles for RB.
B Both chromosomes 13 have the alleles for RB deleted.
C One chromosome 13 carries an allele for RB, but the other chromosome 13 carries the deletion.
D One chromosome 13 carries an allele for RB, but the other chromosome 13 carries a different mutation.
- 22** Which of the following processes could still occur in a chloroplast in the presence of an inhibitor that prevents H^+ from passing through ATP synthase complexes?
- 1 Sugar synthesis
 - 2 Photolysis of water
 - 3 Transfer of electrons down the electron transport chain
 - 4 Oxidation of NADPH
- A** 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

- 23** Some photosynthetic organisms containing chloroplasts that lack PS II (photosystem II) are able to survive. The best way to detect the lack of PS II in these organisms would be to

A test for carbon fixation in the dark.
B test for liberation of oxygen in the light.
C determine the production of starch.
D determine if they have thylakoids in the chloroplasts.

- 24** A mitochondria suspension obtained from liver cells is prepared for investigations of the products of respiration. Acetyl-CoA is added to the suspension.

Which of the following correctly matched the products of glycolysis and Krebs cycle for every oxidation of one glucose molecule in this mitochondria suspension?

A	Product	Glycolysis	Krebs cycle
	ATP	2	2
	Reduced NAD	2	6
	Reduced FAD	0	2
	CO ₂	0	4

B	Product	Glycolysis	Krebs cycle
	ATP	0	2
	Reduced NAD	0	6
	Reduced FAD	0	2
	CO ₂	0	4

C	Product	Glycolysis	Krebs cycle
	ATP	2	4
	Reduced NAD	2	4
	Reduced FAD	0	2
	CO ₂	0	2

D	Product	Glycolysis	Krebs cycle
	ATP	0	4
	Reduced NAD	0	6
	Reduced FAD	0	2
	CO ₂	0	4

- 25** What would be the effect of inhibition of lactate dehydrogenase in a mammalian cell under anaerobic conditions?

A A decrease in glycolysis, due to the lack of NAD.
B A decrease in cell pH, due to the accumulation of lactic acid.
C An increase in ATP production, due to increased amounts of reduced NAD.
D An increase in the activity of Krebs cycle, due to increased amounts of pyruvate.

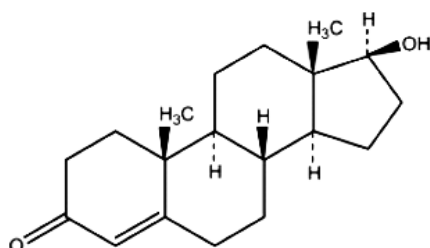
- 26 Tetrodotoxin, a puffer fish toxin, blocks voltage-gated sodium channels. Black widow spider's venom causes the voltage-gated calcium channels to be constantly open. Crotoxin binds irreversibly to acetylcholine receptors. What will happen to the nerve transmission if each toxin is applied?

	Tetrodotoxin	Black widow spider's venom	Crotoxin
A	block action potentials along axon	reduce transmission of impulse across synapse	increase transmission of impulse across synapse
B	increase transmission of impulse across synapse	reduce transmission of impulse across synapse	block action potentials along axon
C	block action potentials along axon	increase transmission of impulse across synapse	reduce transmission of impulse across synapse
D	reduce transmission of impulse across synapse	block action potentials along axon	increase transmission of impulse across synapse

- 27 Which one of the following statements about the transmission across a typical chemical synapse is true?

- A** Neurotransmitter molecules are stored in vesicles in the dendrites.
- B** Vesicles containing neurotransmitter molecules diffuse to the post-synaptic membrane.
- C** The binding of neurotransmitter molecules to receptors transmits an impulse across a synapse.
- D** Action potentials trigger chemical changes that make the neurotransmitter vesicles fuse with the post-synaptic membrane.

- 28 The diagram shows a hormone which affects metabolism in humans.



How does this hormone act on its target cell?

- A** It activates an enzyme cascade that amplifies the hormone signal.
- B** It alters specific receptor sites on the cell surface membrane.
- C** It enters the cell and binds to nuclear receptors.
- D** It inhibits the synthesis of cholesterol molecules.

- 29 Some receptors for hormones like growth factors activate a protein kinase cascade, with the participation of multiple enzymes catalysing multiple steps to effect a change in gene expression. Which of the following statement about a protein kinase cascade are true?
- A Multiple steps in a process allow for the amplification of the signal.
 - B Multiple steps in an activation process means that abnormal stimulation of a cellular response such as growth will not occur even with mutations in more than one gene.
 - C Multiple steps in an activation process ensure that the chemical signal will be sent at the corresponding magnitude as the original hormone signal.
 - D Having multiple steps in the process allows for an “all-or-nothing response”.
- 30 Which of the following statements were **not** part of Darwin’s theory of evolution by natural selection?
- 1 Characteristics acquired during an organism’s lifetime are passed to its offspring
 - 2 Individuals in a sexually reproducing population are different.
 - 3 Organisms produce more offspring than the environment can support.
 - 4 Only those individuals best adapted to the environment survive and reproduce.
 - 5 Regions that encode portions of the polypeptide that are vital for structure and function are less likely to incur mutations.
- A 1 and 2
 - B 1 and 5
 - C 2 and 3
 - D 3 and 5

31 A comparison was made between human, rabbit, mouse and chimpanzee of the

- DNA coding sequence of the β globin gene
- DNA sequence in the introns of the β globin gene
- amino acid sequence of the β globin polypeptide

The data is shown in the table below.

Organisms being compared	Sequence similarity (%)		
	Coding DNA	Intron	Amino acid sequence
Human β globin / chimpanzee β globin	100	98.4	100
Human β globin / rabbit β globin	89.3	67	92.4
Human β globin / mouse β globin	82.1	61	80.1

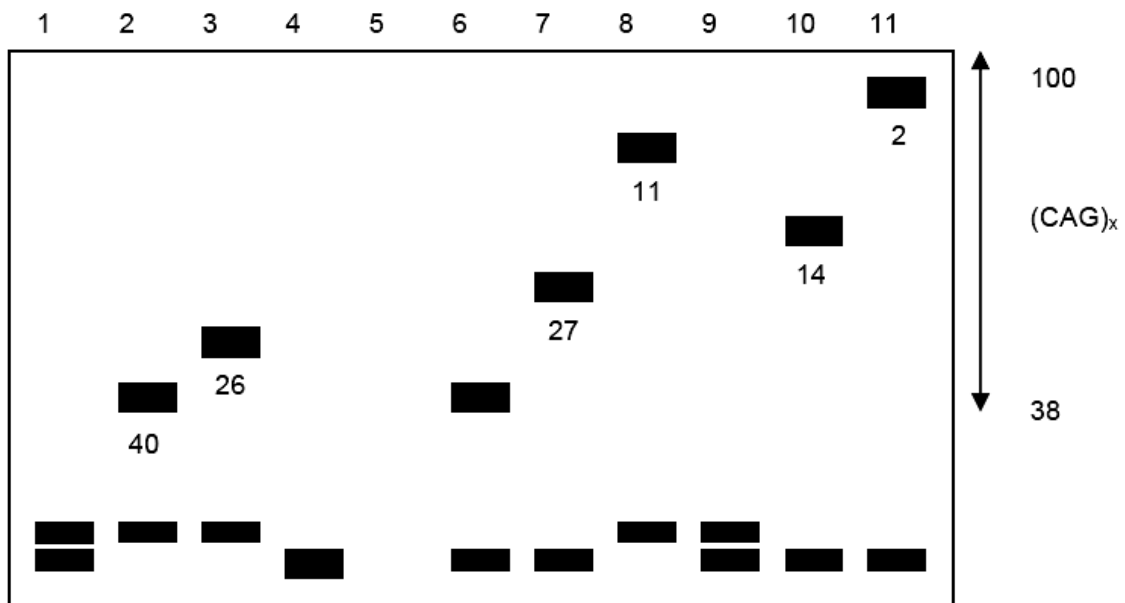
It is possible to conclude from this data that

- A** a human is more closely related to a mouse than to a rabbit.
- B** the variation between chimpanzees and humans occurs in a region of the β globin gene which would code for amino acids.
- C** the variation in the intron sequence between human and mouse would account for some of the differences in the amino acid sequence.
- D** the comparison between rabbit and human indicates that the differences in their DNA did not always make a difference to the amino acid produced.
- 32** A researcher is interested to study the gene expression of a particular ATPase gene in the liver of mice during development using an appropriate DNA library. Which of the following steps are relevant in the creation of this library?
- 1 Restriction digestion of genomic DNA
 - 2 Reverse transcription of mRNA
 - 3 Ligation of single-stranded linkers to the ends of DNA
 - 4 Restriction digestion of plasmid
- A** 1 and 4
- B** 2 and 4
- C** 1, 3, and 4
- D** 2, 3, and 4
- 33** Which of the following is the most powerful way of increasing the specificity of a DNA profile analysis?
- A** Repeat the analysis multiple times.
- B** Increase the number of markers used.
- C** Analyze each marker by PCR rather than RFLP analysis.
- D** Select markers present on the sex chromosomes rather than on the autosomes.

- 34** The data below shows the results of electrophoresis of PCR fragments amplified using primers for the site that has been shown to be altered in Huntington's disease.

The inherited mutation in the Huntington's disease gene abnormally repeats the nucleotide sequence CAG from 36 up to more than 120 times of that. The male parent, shown as individual 2, had the onset of Huntington's disease when he was 40 years old.

Six of his children (individuals **3, 5, 7, 8, 10, 11**) suffer from Huntington's disease, and the age at which the symptoms first began is shown by the number below the band from the PCR fragment.



What is the likely outcome for the normal individuals **4, 6, and 9**?

- A** Individuals **4** and **9** do not have the trait, and will not get Huntington's disease, but individual **6** is likely to start the disease when he reaches his father's age of 40.
- B** Individuals **4, 6, and 9** have not inherited the defect causing Huntington's disease.
- C** Individuals **4, 6, and 9** will still develop Huntington's disease at some point in their lives, since the disease is inherited as a dominant trait.
- D** Two of the three will develop the disease, since it is inherited as a dominant trait, but the data does not allow you to predict which two.

35 Which of the following shows a correct comparison of the various processes?

	Gene amplification	Polymerase chain reaction	DNA replication
A	Occurs in the nucleus	Occurs in the cytoplasm	Occurs in the nucleus
B	Requires nucleic acids	Requires nucleic acid	Does not require nucleic acids
C	Involves hydrogen bonding	Involves hydrogen bonding	Involves hydrogen bonding
D	Does not replicate the entire template strand	Replicates the entire template strand	Replicates the entire template strand

36 Which of the following statements are true about adult stem cells?

- 1 They can undergo self-renewal
- 2 They can be totipotent, pluripotent or multipotent.
- 3 They can differentiate into almost any cell type.
- 4 They can give rise to specialised cells.

- A** 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

37 A patient underwent gene therapy using an adenoviral mediated delivery system. The gene therapy failed and you were asked to find out the cause of the failure. Which of the following could not be the explanation behind the failure in the gene therapy?

- A** Failure of the expressed protein to be folded into the correct conformation.
B The CFTR protein was not expressed in adequate amount.
C Integration of the target gene in the enhancer region.
D Rejection of the vector by the host immune system.

38 What is not expected to be an outcome of the human genome project?

- A** Identification of all genes present in human beings
B Identification of all alleles present in human beings
C Map and sequence the genetic makeup of every human being
D Map and sequence all the genetic material present in the chromosomes of human beings

- 39** Which uses of the information from the human genome project are generally considered to be unethical?
- 1 an insurance company only giving cheap rates to people with genetic predisposition to fewer diseases
 - 2 genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships
 - 3 cytologists developing tests for only some defective genes
 - 4 doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes
 - 5 genetic counsellors giving specific lifestyle information only to people genetically predisposed to risks
 - 6 parents choosing embryos for implantation only after antenatal tests for acceptable genes
- A** 1 and 3
B 1 and 6
C 2 and 5
D 3 and 4
- 40** Which of the following is not a scientific concern relating to creating genetically modified crops?
- A** Commercially important plants become resistant to drought.
B Herbicide resistance may spread to weedy species.
C Non-target insects may be affected.
D Genetically modified plants may lead to unknown risks to human health.